

Sharif University of Technology
Department of Mechanical Engineering

Biomechanical Running Data
Numerical Methods Project

Subject:

Automated Excel Data Analysis, Kinematic Extraction, and
Fatigue Failure Prediction with 3D Structural Validation

Author:

Hasti Shamifard – 401107092

Instructor:

Dr. Adibi

spring 2026

1 Introduction and Problem Statement

In recent years, with the pervasive influence of social networks and fitness-sharing platforms, running has transformed into one of the most widespread healthy lifestyle trends. Fitness influencers and virtual challenges encourage millions of people daily to take up running, constantly highlighting its myriad benefits for cardiovascular health and weight loss. However, a trend that initially emerged to promote public health has now shifted toward extremes; under the influence of social pressure and virtual competition, many amateur runners are quickly pushed to participate in grueling, long-distance events such as marathons and ultra-marathons.

A fundamental and often overlooked question in this wave of enthusiasm is: is this volume of repetitive impact truly beneficial for the structural health of joints, particularly the knee? As the primary shock absorber of the human body, the knee joint endures loads of 2 to 3 times body weight with each running step. While moderate activity helps strengthen cartilage, cyclic and prolonged loading over marathon distances can lead to cartilaginous wear, inflammation, and ultimately, early-onset osteoarthritis. Where is the boundary between “running for health” and “overuse injuries”?

To answer this question, relying on the subjective feelings of runners is insufficient; instead, we require precise, quantitative analysis of the actual forces exerted on the lower extremities under real-world conditions. This study develops a fully automated numerical pipeline in Python to ingest multi-axis raw inertial measurement unit (IMU) data, resolve systematic integration drifts, reconstruct 3D kinematics, and apply solid mechanics principles to evaluate fatigue life consumption of the tibia bone.

2 Methodology and Numerical Implementation

2.1 Signal Unit Mapping and Preprocessing

Raw IMU sensor data logged from wearable sensors (such as ActiGraph devices) are natively stored in G-units ($1\text{ G} = 9.81\text{ m/s}^2$). The pipeline initially converts all raw acceleration components ($a_{x,\text{raw}}, a_{y,\text{raw}}, a_{z,\text{raw}}$) to the SI system:

$$a_x = a_{x,\text{raw}} \cdot g, \quad a_y = a_{y,\text{raw}} \cdot g, \quad a_z = a_{z,\text{raw}} \cdot g \quad (1)$$

A 4th-order zero-phase Butterworth bandpass filter is then applied with cutoffs adjusted at 0.15 Hz (low-frequency) to preserve slow athletic foot translations, and 45 Hz (high-frequency) to retain heel-strike transient shocks.

2.2 3D Numerical Integration and Zero Velocity Update (ZUPT)

A common pitfall in lower-extremity kinematics is integrating the scalar acceleration magnitude directly. Since the magnitude $a_{\text{mag}}(t) = \sqrt{a_x(t)^2 + a_y(t)^2 + a_z(t)^2}$ is strictly non-negative, any sensor noise or multi-directional motion yields a strictly positive accumulation of error. Mathematically, integrating a non-negative scalar function means velocity can never decrease during deceleration phases, representing a physical impossibility where the foot constantly accelerates.

To preserve physical reality, the velocity must be reconstructed as a vector. Acceleration components are projected onto the local coordinate frame and integrated individually

using the Forward Euler method:

$$v_x[i] = v_x[i - 1] + a_x[i] \cdot \Delta t \quad (2)$$

$$v_y[i] = v_y[i - 1] + a_y[i] \cdot \Delta t \quad (3)$$

$$v_z[i] = v_z[i - 1] + a_z[i] \cdot \Delta t \quad (4)$$

Integrating the components individually allows the reconstructed velocity to naturally decrease when acceleration components are negative (e.g., during braking or deceleration phases).

However, due to inherent sensor biases and high-frequency noise, integrating raw acceleration introduces a linear drift in velocity and quadratic drift in displacement over time. To constrain this drift, we implement the Zero Velocity Update (ZUPT) algorithm. ZUPT exploits a key biomechanical constraint of the human gait cycle: during the stance phase (when the foot is flat on the ground), the physical velocity of the foot relative to the earth is exactly zero.

The stance phase is detected using an energy-based threshold on the dynamic acceleration magnitude:

$$\text{Stance State}[i] = \begin{cases} \text{True}, & \text{if } a_{\text{mag}}[i] < \text{Threshold}_{\text{ZUPT}} \\ \text{False}, & \text{otherwise} \end{cases} \quad (5)$$

where $\text{Threshold}_{\text{ZUPT}} = 0.6 \text{ m/s}^2$. To prevent false positives from transient zero-crossings, a minimum stance duration window (≈ 5 consecutive samples or 0.05 s) is enforced. When a stance state is confirmed, the velocity vector is forced to zero, resetting the drift accumulation:

$$\mathbf{v}[i] = [v_x[i], v_y[i], v_z[i]]^T = [0.0, 0.0, 0.0]^T \quad (6)$$

Finally, the actual speed scalar $v[i]$ is calculated as the magnitude of the corrected velocity vector:

$$v[i] = \sqrt{v_x[i]^2 + v_y[i]^2 + v_z[i]^2} \quad (7)$$

2.3 Gait Segmentation and Running Phase Isolation

Real-world athletic trials rarely consist of uniform, uninterrupted running. The 20-minute IMU dataset captured in this study represents a dynamic, multi-modal exercise session containing intervals of standing, warm-up walking, and active running. Including all these heterogeneous activities in a singular mechanical stress analysis would severely underestimate the running-specific cyclic load and distort the fatigue predictions. Walking and standing involve low-impact, quasi-static forces ($F_{\text{GRF}} \approx 1.0\text{--}1.2 \text{ BW}$) that fall well below the fatigue damage threshold of cortical bone. In contrast, running induces high-impact, cyclic loads ($F_{\text{GRF}} \geq 2.5 \text{ BW}$) which are the primary drivers of bone micro-damage and stress fractures.

Consequently, the pipeline must segment the continuous trial and isolate the active “Running” intervals. Classification is performed using physical speed thresholds applied to a smoothed speed profile $v_{\text{smooth}}[i]$. During running, the instantaneous velocity of the ankle fluctuates violently within each individual stride due to rapid acceleration during the swing phase and sudden deceleration upon ground impact. To analyze the macroscopic forward progression of the athlete rather than these micro-oscillations, a smoothing filter must be applied. Conceptually, this process is highly analogous to a vehicle’s speedometer:

rather than displaying the rapid, instantaneous vibrations of the tires, the speedometer displays a steady, averaged velocity of the vehicle’s forward translation. In this pipeline, we compute a 2.0-second symmetric moving average window to filter out high-frequency intra-stride oscillations and expose the underlying gait pace:

$$v_{\text{smooth}}[i] = \frac{1}{2N+1} \sum_{k=-N}^N v[i+k], \quad N = \frac{1.0 \text{ s}}{\Delta t} \quad (8)$$

Based on the smoothed velocity, each sample is classified into one of three distinct gait phases:

$$\text{Gait}[i] = \begin{cases} \text{Standing,} & \text{if } v_{\text{smooth}}[i] < 0.5 \text{ m/s} \\ \text{Walking,} & \text{if } 0.5 \text{ m/s} \leq v_{\text{smooth}}[i] \leq 1.1 \text{ m/s} \\ \text{Running,} & \text{if } v_{\text{smooth}}[i] > 1.1 \text{ m/s} \end{cases} \quad (9)$$

To eliminate rapid, non-physical state switching (flickering) caused by transient acceleration spikes, a temporal smoothing pass is executed. Any isolated gait phase segment lasting less than 2.0 seconds is merged into its neighboring dominant state. Once the continuous signal is segmented, only the classified “Running” intervals are extracted and forwarded to the kinematic and kinetic analytics block. This isolation ensures that the subsequent marathon simulation and tibia fatigue calculations are parameterized exclusively by steady-state running mechanics.

2.4 Biomechanical Kinetic Force Correction

Accelerometers mounted on the ankle record extreme transient impact accelerations. Applying Newton’s second law directly to local foot acceleration yields unphysical loads. To model the shock-absorbing mechanism of the lower extremity joints (knee, hip, and ankle), we introduce a joint damping/attenuation coefficient ($\eta = 0.25$) to scale the foot’s peak dynamic acceleration down to the center of mass (CoM) equivalent acceleration:

$$F_{\text{GRF}} = m \cdot (\eta \cdot a_{\text{mag_seg}}[hs_idx] + g) \quad (10)$$

For a 70 kg athlete, this yields realistic Ground Reaction Force (GRF) peaks averaging between 1700 N and 2300 N during steady running.

3 Experimental Results and Analytics

The optimized numerical pipeline was executed on a 20-minute experimental IMU dataset consisting of 119,900 samples logged at a sampling rate of 100 Hz. This section presents a detailed, quantitative analysis of the extracted gait characteristics, running intervals, and biological fatigue indicators.

3.1 Gait Phase Distribution

The automated gait classification algorithm segmented the continuous 20-minute trial into three fundamental gait phases based on the smoothed velocity profile:

- **Standing:** 52,534 samples (8.8 minutes, representing 43.81% of the trial).
- **Walking:** 25,599 samples (4.3 minutes, representing 21.35% of the trial).

- **Running:** 41,767 samples (7.0 minutes, representing 34.84% of the trial).

This temporal division indicates that the exercise session was not a continuous run but rather an interval-based training profile, containing standing breaks and warm-up/cool-down walking periods.

3.2 Detailed Running Intervals

A total of 12 distinct running intervals were successfully segmented. Table 1 provides a comprehensive overview of the kinematics (duration, speeds) and kinetics (forces, heel strikes) computed for each isolated interval.

Table 1: Kinematic and kinetic metrics calculated for each of the 12 isolated running intervals.

Int.	t_{start} (s)	t_{end} (s)	Dur. (s)	v_{mean} (m/s)	v_{max} (m/s)	v_{SD} (m/s)	HS	F_{max} (N)	F_{mean} (N)	F_{SD} (N)
1	1.6	4.9	3.3	1.119	2.353	0.545	3	963.6	890.2	61.4
2	44.3	55.2	10.9	1.262	3.729	0.819	13	3850.8	1798.5	926.3
3	151.2	157.2	6.0	1.321	2.979	0.471	13	3952.6	1852.0	882.7
4	371.3	373.7	2.4	1.771	3.447	1.043	0	nan	nan	nan
5	429.6	625.0	195.4	1.475	3.838	0.599	449	4201.8	2183.9	561.4
6	820.8	868.0	47.2	1.422	3.455	0.574	81	2744.7	1814.6	494.3
7	902.2	904.3	2.1	1.333	2.470	0.490	3	1812.4	1701.3	151.4
8	917.2	919.5	2.3	1.093	2.320	0.652	3	1794.0	1337.8	388.0
9	926.6	931.9	5.2	1.289	3.348	0.743	7	1717.6	1481.1	233.2
10	943.7	1008.7	65.0	1.783	4.797	0.751	144	4351.9	2269.7	734.2
11	1023.1	1088.9	65.7	1.716	4.256	0.716	136	3820.4	2188.9	609.7
12	1093.5	1105.6	12.1	2.111	5.474	1.365	4	1384.2	1016.1	222.1

3.2.1 Detailed Discussion of Table 1:

Several critical biomechanical trends can be observed from the segmented intervals:

- **Transient Intervals and Algorithm Edge Cases:** In extremely short intervals, such as Interval 4 (duration of 2.4 seconds), zero heel strikes (HS) were registered, resulting in undefined (*nan*) force parameters. This is a realistic outcome representing rapid transitional accelerations. The runner briefly accelerated past the running threshold (1.1 m/s) but did not complete a full, rhythmic stride cycle on the monitored foot within that narrow window.
- **Steady-State Run vs. Sprinting:** Interval 5 acts as the primary steady-state running session, spanning over 3.25 minutes with a stable speed profile ($v_{\text{mean}} = 1.475$ m/s, $v_{\text{SD}} = 0.599$ m/s). Conversely, Interval 12 represents a terminal sprint, achieving the highest maximum speed of 5.474 m/s (approximately 19.7 km/h) and a high velocity standard deviation (1.365 m/s), showing intense acceleration and deceleration.

3.3 Combined Running Summary

To evaluate the cumulative biomechanical exposure, all 12 running intervals were combined. The total combined active running duration is 6.96 minutes, during which a total of 856 valid heel strikes were detected on the monitored foot. Table 2 compiles the global statistics for running velocity and ground reaction forces.

Table 2: Global velocity and kinetic metrics across all combined running intervals.

Metric	Mean	Min	Max
Speed mean (m/s)	1.47	1.09	2.11
Speed max (m/s)	3.54	2.32	5.47
Speed SD (m/s)	0.73	0.47	1.36
Force max (N)	2781.28	963.62	4351.94
Force min (N)	932.71	810.32	1487.27
Force SD (N)	478.62	61.45	926.27
Force mean (N)	1684.90	890.23	2269.70

3.3.1 Detailed Discussion of Table 2:

- **Kinematic Profile:** The global average speed of 1.47 m/s indicates a light jog, while the average of the maximum speeds reaches 3.54 m/s, confirming that the athlete regularly incorporated high-intensity intervals.
- **Kinetic Profile and Weight Ratio:** Under static conditions, the 70 kg runner exerts a gravitational force of:

$$F_{\text{gravity}} = 70.0 \text{ kg} \times 9.81 \text{ m/s}^2 = 686.7 \text{ N} \quad (11)$$

Comparing this to our kinetic summary, the global mean ground reaction force ($F_{\text{mean}} = 1684.90 \text{ N}$) corresponds to exactly **2.45** times the body weight (2.45 BW). This active force profile is exceptionally accurate, perfectly aligning with classic biomechanics literature which places normal running impact forces between 2.2 and 2.5 body weights. Furthermore, the maximum transient force registered reaches 4351.94 N (6.34 BW), illustrating the severe shock loads absorbed during sprinting.

3.4 Dynamic Fatigue Trend Assessment

To evaluate how the athlete's motor control and impact mitigation changed over the course of the 20-minute session, the pipeline compared running parameters between the first quarter (early phase) and the last quarter (late phase) of the running intervals. Table 3 presents the comparative results.

Table 3: Comparative analysis of early vs. late running intervals to assess athletic fatigue.

Metric	Early Phase	Late Phase	Change (Δ)	Status
Force SD (N)	493.86	415.92	-15.8%	Stable
Force mean (N)	1344.35	1602.47	+19.2%	Fatigue Detected
Speed SD (m/s)	0.61	0.94	+54.3%	Fatigue Detected

3.4.1 Detailed Discussion of Table 3:

The comparative trends reveal a clear biomechanical signature of physiological fatigue:

- **Reduced Muscle Shock Absorption (+19.2% in F_{mean}):** In the early stages of running, the athlete's average impact force was relatively low (1344.35 N $\approx$ 1.96 BW). However, by the late stages, this average force rose significantly to 1602.47 N $\approx$ 2.33 BW, representing a 19.2% increase. Under prolonged exertion, the lower-limb musculature (primarily the quadriceps, gastrocnemius, and soleus) becomes exhausted, losing its active, contractile energy dissipation capability. As a result, the damping system fails, and the runner experiences significantly harder landings, transferring higher mechanical stress directly to passive skeletal structures (tibia and knee cartilage).
- **Loss of Rhythmic Gait Control (+54.3% in Speed SD):** The variability of speed (represented by Speed SD) escalated dramatically by 54.3%, rising from 0.61 m/s to 0.94 m/s. While fresh runners exhibit a highly rhythmic and steady-state pace, neuromuscular fatigue compromises motor unit recruitment and joint coordination. This leads to gait instability, characterized by irregular stride speeds, struggling to maintain pace, and a fluctuating velocity profile.

4 Marathon Simulation and Bone Failure Analysis

Using the steady-state athletic parameters extracted from Interval 5 ($v_{\text{mean}} = 1.475$ m/s and a stride frequency of 1.5 Hz under a sustained marathon distance $D = 42,195$ m):

4.1 Kinematic and Step Estimation

- **Total Marathon Time:** $T_{\text{total}} = \frac{42195}{1.475} \approx 28,606$ s $\approx$ 7h 56min
- **Total Stride Count (One Foot):** $n = 1.5 \text{ Hz} \cdot 28,606 \text{ s} \approx 42,909$ steps

4.2 Cumulative Kinetic Load

The total cumulative mechanical force absorbed by a single ankle joint during the marathon is calculated as:

$$\text{Cumulative Force} = n \cdot F_{\text{mean}} \approx 42,900 \cdot 1684.9 \text{ N} \approx 72.3 \text{ MN} \quad (12)$$

This is equivalent to dissipating a cumulative weight of approximately 7,370 Tons under standard gravity.

4.3 Stress Modeling and Miner's Rule

Assuming a Tibia bone cross-sectional area of $A \approx 4 \times 10^{-4} \text{ m}^2$, the stress amplitude under cyclic bending moments is modeled at $\sigma_a \approx 30 \text{ MPa}$. Using the empirical S-N relationship for human cortical bone:

$$\log_{10}(N_f) = 11 - 0.18 \cdot \sigma_a \quad (13)$$

For $\sigma_a = 30 \text{ MPa}$, the cycles to fatigue failure is $N_f = 10^{5.6} \approx 398,107$. The cumulative Damage Index (D) is:

$$D = \frac{n}{N_f} = \frac{42,909}{398,107} \approx 0.1078 \quad (14)$$

This indicates that approximately 10.8% of the bone’s structural fatigue life is consumed in a single marathon under active mechanical fatigue. This finding carries significant clinical and biomechanical implications:

- **Absence of Real-Time Remodeling:** While living bone is dynamic and governed by Wolff’s Law, the biological bone remodeling process (initiated by osteoclastic resorption followed by osteoblastic bone deposition) operates on a slow temporal scale of weeks to months. Consequently, during the 4 to 8 hours of continuous mechanical loading during a marathon, no active biological healing can occur. The 10.8% calculated damage represents a net, unmitigated accumulation of physical microcracks within the cortical extracellular matrix by the end of the race.
- **Stress Fracture Cascade:** Under Miner’s linear cumulative damage framework, if a runner fails to allow for a sufficient post-race recovery window, subsequent training sessions will accumulate damage on top of this initial baseline. If the rate of mechanical microdamage propagation outpaces the biological rate of osteonal remodeling, these microscopic matrix cracks will coalesce, ultimately propagating into macroscopic stress fractures—a classic overuse injury prevalent in endurance athletes.

5 Conclusion

This study presented a revised numerical and physical pipeline to analyze raw acceleration data. By introducing 3D vector integration and joint shock absorption attenuation, the calculated kinematics and kinetics were validated against standard biomechanical limits. The fatigue assessment highlighted a 19.2% rise in impact forces and a 54.3% increase in gait variability as fatigue progressed, demonstrating that overuse during extreme runs significantly accelerates damage index accumulation (10.8% per marathon). This highlights the critical need for scheduled recovery periods to allow biological bone remodeling to outpace mechanical damage propagation.

References

- [1] Bennett, B. (2018). *IMU data on Walking and Running (Unlabeled)*. California State University, East Bay. Kaggle Dataset. <https://www.kaggle.com/datasets/aroor/imu-data-on-walking-and-running-unlabeled>
- [2] Wolff, J. (1986). *The Law of Bone Remodelling*. Springer-Verlag, Berlin Heidelberg. (Original work published 1892).
- [3] Miner, M. A. (1945). Cumulative damage in fatigue. *Journal of Applied Mechanics*, 12(3), A159–A164.
- [4] Skog, I., Handel, P., Nilsson, J. O., & Rantakokko, J. (2010). Zero-velocity detection—An algorithm evaluation. *IEEE Transactions on Biomedical Engineering*, 57(11), 2650–2656.
- [5] Carter, D. R., & Caler, W. E. (1983). Cycle-dependent and time-dependent bone fracture with repeated loading. *Journal of Biomechanical Engineering*, 105(2), 166–170.